Rajarata University of Sri Lanka

Department of Computing

COM 3401 – Data Communication and Networking
Laboratory 03 – IPv4 addressing and subnetting
Time: 2 Hours

Intended Learning Outcomes (ILOs)

By the end of this practical, students should be able to:

1. Learning on IPv4 addressing and to identify network and host portions
2. Getting familiar with subnetting a configuring network based on requirements

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Pre-Practical Preparation

1. **Reading/Resources**: Refer to the topics covered in Lecture 3 and Lecture 4

Practical Objective:

Explain how network and host portions of an IPv4 address paly a vital role in addressing. Making students aware of calculations related to these address portions and determine network/broadcast addresses of each subnet. Students will learn how to subnet a network based on the requirements of the scenario specially through the usage of VLSM.

Practical Activity

Activity 1:

Subnetting in LAN

- Objectives: Calculating valid address portions in a subnet and configuring LANs
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x switch (2960)
        - 4 x PC
    3. Connect PCs to the switch using Copper straight-through cables and name it as LAN1
    4. Assign the subnet 192.168.10.0/26 to LAN

5. Ping PC3 from PC0
6. Randomly ping between available endpoints to verify the connectivity

Activity 2:

Subnetting in LAN

- Objectives: Calculating valid address portions in a subnet and configuring LANs
- Instructions:

1. Create a new project
    2. Add following devices:
        - 2 x switch (2960)
        - 6 x PC
    3. Connect PC0, PC1 and PC2 to the switch0 using Copper straight-through cables and name it as LAN1
    4. Connect PC3, PC4 and PC5 to the switch1 using Copper straight-through cables and name it as LAN2
    5. Connect two switches using Copper straight-through cables
    6. Assign the subnet 192.168.10.0/26 to LAN1
    7. Assign the subnet 192.168.10.64/26 to LAN2
    8. Assign the appropriate subnet mask manually to all PCs
    9. Verify the inter-network connectivity and intra-network connectivity

Activity 3:

Introduction to subnetting and VLSM

- Objective: Making students aware on how to do calculations regarding subnetting how to apply VLSM
- Instructions:

    1. Go through the given scenario

        A company has the network 192.168.50.0/24 and requires two departments.

Department	Hosts required
Sales	40
HR	12

Requirements:

2. Calculate the network addresses
3. Calculate the broadcast addresses
4. Identify valid host range
5. Build the subnets
6. Configure PCs
7. Test communication within each LAN

Activity 4:

Subnetting simulation for a real-world scenario using VLSM

- Objective: Making students aware on how subnetting play a vital role in real-world scenario
- Instructions:
  1. Go through the given scenario

     A company has the network 192.168.20.0/24 and requires four departments.

Department	Hosts required
Administration	30
Finance	25
HR	20
IT	50

Requirements:

2. Calculate the network addresses
3. Calculate the broadcast addresses
4. Identify valid host range
5. Build the topology
6. Configure PCs
7. Test communication within each LAN